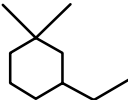
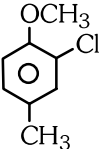
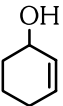
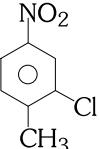
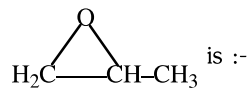


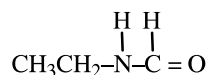
NOMENCLATURE & ISOMERISM

1.  has the IUPAC name—
 (1) 1-Ethyl-3,3-dimethyl cyclohexane
 (2) 3-Ethyl-1,3-dimethyl cyclohexane
 (3) 3-Ethyl-1,1-dimethyl cyclohexane
 (4) 3,3-Dimethyl-1-ethyl cyclohexane
2.  is known as—
 (1) 4-Chloro-2-methyl anisole
 (2) 2-Chloro-4-methyl anisole
 (3) 3-Chloro-4-methyl anisole
 (4) 1-Chloro-3-methyl anisole
3. $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_2 - \underset{\text{CH}_3}{\text{CH}} -$ Radical has IUPAC name—
 (1) 4-Methyl pentyl (2) 1,3-Dimethyl butyl
 (3) 2,4-Dimethyl butyl (4) 3-Methyl pentyl
4.  has the IUPAC name—
 (1) Cyclohex-1-en-3-ol
 (2) Cyclohex-1-en-6-ol
 (3) Cyclohex-2-en-1-ol
 (4) Cyclohex-5-en-1-ol
5. $\text{BrCH}_2 - \text{CH} = \text{CH}_2$ has the IUPAC name—
 (1) 3-Bromo prop-1-ene (2) 1-Bromo prop-2-ene
 (3) 1-Bromo prop-1-ene (4) 2-Bromo prop-2-ene
6.  has the IUPAC name—
 (1) 4-Methyl-5-Chloro-1-nitrobenzene
 (2) 1-Methyl-4-nitro-6-Chloro benzene
 (3) 2-Chloro-1-methyl-4-nitrobenzene
 (4) 1-Chloro-2-methyl-5-nitrobenzene

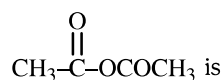
7. The IUPAC name for the compound



- (1) Propylene oxide (2) 1,2-Oxopropane
 (3) 1,2-Epoxypropane (4) 1,2-Propoxide
8. 3-Phenylprop-2-enoic acid is IUPAC name of
 (1) Mendallic acid
 (2) Pivallic acid
 (3) Succinic acid
 (4) Cinnamic acid
9. The correct IUPAC name of
 $\text{CH}_3 - \underset{\text{C}_2\text{H}_5}{\text{C}} = \underset{\text{C}_2\text{H}_5}{\text{C}} - \text{OCH}_3$
 (1) 1,2-Diethylbutane
 (2) 2-Ethyl-3-methylpentane
 (3) 3,4-Dimethyl-hex-3-ene
 (4) 3-Methoxy-4-methylhex-3-ene
10. One among the following is the correct IUPAC name for the compound

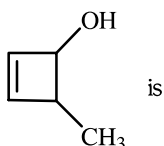


- (1) N-Formylaminoethane
 (2) N-Ethylformylamine
 (3) N-Ethylmethanamide
 (4) Ethylaminomethanal
11. The correct IUPAC name of



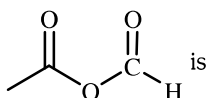
- (1) Methyl ethanoate (2) Acetato ethanoate
 (3) Ethanoic anhydride (4) Ethanoyl ethanoate
12. The correct IUPAC name of
 $\text{HOOC} - \underset{\text{COOH}}{\text{CH}} - \text{COOH}$ is
 (1) Tricarboxymethane
 (2) Propanetricarboxylic acid
 (3) Tributanoic acid
 (4) Methanetricarboxylic acid

13. The IUPAC name of



is

- (1) 3-Methylcyclobut-1-en-2-ol
 (2) 4-Methylcyclobut-2-en-1-ol
 (3) 4-Methylcyclobut-1-en-3-ol
 (4) 2-Methylcyclobut-3-en-1-ol
14. The IUPAC name of the compound
-
- is
- (1) 4-Methylcyclopent-1-en-2-ol
 (2) 2-Methylcyclopent-4-en-1-ol
 (3) 3-Methylcyclopent-1-en-2-ol
 (4) 5-Methylcyclopent-2-en-1-ol
15. The IUPAC name of the hydrocarbon
 $\text{CH} \equiv \text{C} - \text{CH} = \text{CH} - \text{CH} = \text{CH}_2$ is
- (1) Hexa-3, 5-dien-5-yne
 (2) Hexa-1, 2-dien-1-yne
 (3) Hexa-1, 3-dien-5-yne
 (4) Hexa-3, 5-dien-1-yne
16. The correct IUPAC name of



is

- (1) Butane-2, 4-dione
 (2) Formyl ethanoate
 (3) Acetic anhydride
 (4) Ethanoicmethanoic anhydride
17. The correct IUPAC name of the
-
- is
- (1) 3-Aldo-2-oxopropanoic acid
 (2) 2,3-Dioxopropanoic acid
 (3) 1-Hydroxy propane-1, 2, 3-trione
 (4) 2-Aldo-2-keto methanoic acid

18. Including cyclic structures, how many structural isomers are possible for the formula C_4H_8 ?

- (1) 3 (2) 5
 (3) 2 (4) 4

19. The number of isomeric aldehydes and ketones with formula $\text{C}_5\text{H}_{10}\text{O}$ are (structural isomers only)

- (1) 7 (2) 6
 (3) 5 (4) 8

20. Which of the following isomerism is shown by ethyl acetoacetate ?

- (1) Enantiomerism
 (2) Keto-enol isomerism
 (3) Geometrical isomerism
 (4) Diastereoisomerism

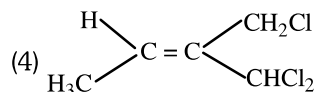
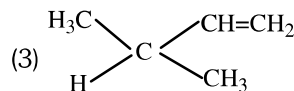
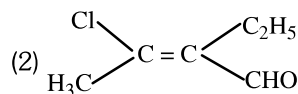
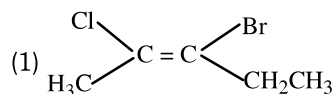
21. $\text{C}_4\text{H}_4\text{O}_4$ can be

- (1) A cyclic ester of dibasic acid
 (2) A cis-dibasic acid
 (3) A trans-dibasic acid
 (4) All are true

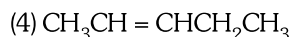
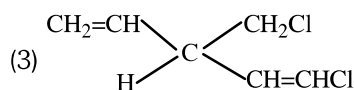
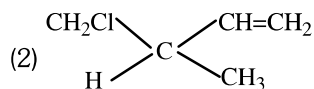
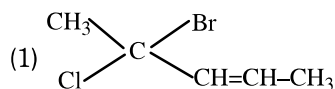
22. The total number of structural isomers of $\text{C}_4\text{H}_{10}\text{O}$ will be

- (1) 4 (2) 5
 (3) 6 (4) 7

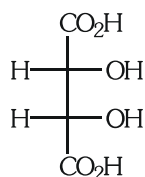
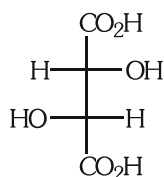
23. The E-isomer amongst the following is



24. One among the following will not show geometrical isomerism, identify that compound.



25. Which of the following statement is correct ?
- The presence of chiral carbon is essential condition for enantiomerism
 - Functional isomerism is a kind of stereoisomerism
 - The compounds containing one chiral carbon are always optically active
 - All the statements are wrong
26. The two isomers given below are



- Enantiomers
 - Diastereomers
 - Mesomers
 - Position isomers
27. Which of the following compound could be optically active?
- 4-Chloro-1-hydroxybutane
 - Tertiary butyl alcohol
 - Secondary butyl amine
 - n-Butyl alcohol

28. The optical inactivity of meso-tartaric acid is because of

- absence of chiral carbon atoms
- external compensation
- internal compensation
- presence of asymmetrical carbon atoms

29. How many chiral carbons are present in glucose molecule $\text{CHO}(\text{CHOH})_4\text{CH}_2\text{OH}$?

- 4
- 3
- 2
- 1

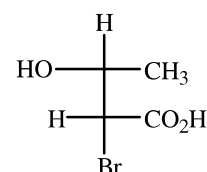
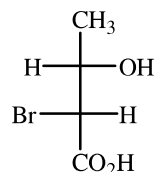
30. Which of the following will not be able to show optical isomerism?

- 1,2-Propadiene
- 2,3-Pentadiene
- sec-Butyl alcohol
- 2, 3-Butanediol

31. Glucose and Fructose are

- Chain isomers
- Position isomers
- Functional isomers
- Optical isomers

32. The molecules represented by the following two structures are



- Identical
- Enantiomers
- Diastereomers
- Epimers